

Terminating Installed Communication State When the Control Path Dies: Deadline Expiry and Delivery-Bounded Configuration Leases for Intermittent Pre-Disaster Monitoring

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Abstract—When a mountain-site backhaul fails, the network keeps executing communication state that was installed before the failure. Unacknowledged records keep filling the few recovery access batches, and a dense sampling configuration that can no longer be countermanded keeps draining a battery overnight. We study where, and on what evidence, these two forms of *persistent installed state* can be terminated, for a battery-plus-solar LoRaWAN Class A deployment with an autonomous gateway, a cellular primary and a rate-limited BeiDou short-message backup. One decision problem—local termination under partial evidence—covers both: a node holding only its clock, battery, cache, recent access receipts and install-time bounds decides to retain or release a record and to hold or relax a configuration, under no-false-release, energy-survival, locality and no-false-completion constraints. A record’s termination time is its business deadline, locally certain; the rule is bit-identical to standard Bundle-Protocol per-record expiry, and its *cross-segment placement* decides the gain: source-local expiry raises outage on-time delivery $202 \rightarrow 434$ and 2.52-fold over ten seeds, whereas suppressing expired sends at the gateway alone back-pressures access (-8). A configuration’s termination time is not locally observable, because it depends on mission revisions travelling the failed path and on return-slot geometry held at the gateway. A clock night-guard suffices when the outage aligns with sunset and energy is ample; under overcast skies it leaves 20–31 nodes to discharge fatally, and a fixed config TTL short enough to save them falsely releases 183 still-deliverable obligations in another outage phase. A *delivery-bounded lease*—an ordinary lease whose bound is the last time the current profile’s data can still meet an obligation—removes those deaths in every phase without false release. The mechanisms are ordinary primitives with a re-derived bound. A formative LLM study locates the interface fault they target: LoRa access receipts read as backhaul state. All claims come from deterministic, repository-reproducible runs; end-to-end and cross-model agent validation is future work.

Index Terms—Persistent installed state, intermittent backhaul, LoRaWAN Class A, satellite short message, deadline expiry, configuration lease, partial observability, pre-disaster monitoring.

I. INTRODUCTION

Consider a field node running an authorised dense sampling profile. While the cellular backhaul is up, the centre sends the configuration and the node applies it at its next LoRaWAN

Class A receive window. The backhaul then fails for sixteen hours, and three things keep happening with no control path to stop them. The node keeps sampling and transmitting at the dense rate, drawing energy the battery may not have through the night, while a downgrade authored after the failure cannot reach it. Records already cached at the node are, under the standard hold-until-acknowledged edge discipline, retransmitted oldest-first on every narrow access opportunity, including records whose business deadline has passed. The centre and gateway cannot tell whether missing data were never sampled, sit at a node, reached the gateway, or merely lack a return slot, nor whether a command is queued, in transit or applied. The network is not stuck: *installed state is still executing*, spending energy, access slots and metered short-message bytes on work that can no longer meet any obligation.

This is a communication-systems failure with a specific shape: **the termination of a persistent installed state depends on a control/acknowledgement path that has itself failed**. SateRIoT observes that lossy satellite windows waste COTS-radio retransmissions and redesigns link estimation and multi-hop sharing around that measured behaviour [1]; semantic HARQ reuses the mature HARQ primitive but changes its error criterion to match a new communication object [2]. In the same spirit we ask which ordinary mechanism already terminates each kind of installed state, and where a genuine gap remains once the ordinary mechanisms are applied fairly.

A. Two kinds of persistent installed state

A **record state** is an unacknowledged sample in the node cache, occupying a slot in every future access batch until acknowledged or discarded. A **configuration state** is the node’s current sampling/reporting profile, installed by an applied command, which drives energy and airtime until a newer command or a local condition ends it. Both are created while connected and both outlive the connection. They differ in what evidence terminates them: a record’s useful lifetime ends at an absolute business deadline the source computes from the public mission cadence, whereas a configuration’s useful lifetime ends when the data it produces can no longer be

delivered—a fact depending on mission *revisions*, which reach the field over the failed backhaul, and on the geometry of future return opportunities, which the gateway and not the node holds. This asymmetry, made precise in §IV, is the paper’s technical centre.

B. What strong ordinary mechanisms already settle

We first assemble deliberately strong baselines with the same legal information: energy-aware feed-forward sampling, an omniscient offline packer, rolling-horizon MPC, max-coverage/recency backup packing, earliest-deadline-first and obligation-greedy queues, piecewise non-prescient mission views, and symmetric resource relaxations. Over the policy families tested, re-optimising collection, scheduling or packing under fixed resources yields no robust on-time-delivery gain; the large fixed-resource gains require added backhaul capacity or energy. We report this as a scoped negative result. Two ordinary termination mechanisms also work well: per-record expiry, where releasing a cached record at its business deadline is standard Bundle-Protocol lifetime management [3], to which a prior “certificate-driven purge” proves bit-identical; and a local clock/energy guard, where a node-local rule relaxing dense sampling at dusk removes overnight deaths whenever the outage aligns with sunset and the battery is adequate. We credit both as mature primitives.

C. The residual

Two controlled experiments isolate what ordinary mechanisms do *not* cover. Releasing expired records at the *gateway*—stop sending, leave the source cache untouched—withholds the acknowledgement, so the expired record keeps filling the node’s access batch and on-time delivery moves *down* 202 → 194; the disposition must be consistent across the access and backhaul segments, which single-hop AoI discard and hop-local bundle expiry do not address. And the clock guard acts only at dusk: when an authorised daytime downgrade falls inside an outage, the node runs dense for hours after its data ceased to be deliverable, which under overcast skies is fatal (20–31 nodes), while a naive short config TTL that relaxes early enough to save those nodes falsely releases obligations still deliverable over the backup in a differently phased outage (183 lost). No fixed TTL is safe and sufficient across phases. The residual is therefore not “terminate state”—ordinary expiry and guards do that—but **computing the configuration’s termination bound from delivery geometry the node does not have, and installing it while connected so it can be enforced locally while disconnected.**

D. Contributions

- 1) **One termination problem for persistent installed state (§III, §IV).** Records and configurations are modelled as state installed while connected and executed while disconnected, each state’s release time is derived from the obligation chain, and a single local decision problem is stated under four

constraints—no-false-release, energy survival, evidence locality, no-false-completion—with a three-class confirmed/unconfirmed/unreachable discipline that makes *unknown* a first-class outcome.

- 2) **Record termination is standard expiry; cross-segment placement is the gain (§V-A, §VII-B).** Deadline-aligned source expiry is bit-identical to BPv7 per-record lifetime on ten seeds. The system contribution is an end-to-end business-deadline basis with source/gateway consistency, which doubles outage on-time delivery and drives expired backup records to one, while gateway-only suppression moves cost to the access segment. Full-horizon time-aware attribution corrects an earlier accounting that had labelled 830 late-access failures as backhaul-capacity failures.
- 3) **A delivery-bounded configuration lease for the residual (§V-B, §VII-C).** The ordinary config-TTL/lease primitive keeps its form; its fixed-duration bound is replaced by the last deliverable time of the current profile’s data, computed at the gateway while connected and enforced at the node with no downlink. Across outage phases and harvest levels it removes the 20–31 overcast deaths the clock guard leaves and avoids the 183-obligation false release of the best fixed TTL, while in ample-energy regimes it only removes wasted dense work.
- 4) **A formative real-agent study that locates the interface fault (§VII-F).** Symmetric accounting of 1089 live decisions shows LoRa access receipts read as backhaul state in both directions; an unknown-aware projector removes the healthy-link false assertions in offline replay. The agent is a *controller candidate*, not part of the problem definition.

We claim no LLM advantage over a solver, no new discard algorithm, and no global scheduling upper bound. The deterministic mechanisms are in-layer tools available equally to rules, solvers and agents.

II. SCENARIO AND OPERATING POINT

Fourteen sensor nodes in two groups monitor displacement and rainfall on a slope. Nodes sample and upload to a local gateway over LoRaWAN Class A; the gateway forwards to a central server over cellular backhaul, with a BeiDou short-message service as a sparse, metered backup carrying field-to-centre messages only, never downlink configuration. Nodes are battery plus small solar, the mission is pre-disaster and routine with authorised warning-level changes, and no component judges landslide risk. Node count, role mapping and disturbance parameters follow the frozen Task v1.1 instance manifest; an earlier 16-node sketch is not used as a field fact.

Authority comes from two Chinese geological-hazard monitoring standards, DZ/T 0460–2023 and DZ/T 0450–2023, retrieved and clause-audited in the repository [4], [5]. §5.3.3 permits remote dynamic adjustment of sampling and upload frequency by warning level and gives no numeric levels, so the periods used here are declared research choices; §8.4.1 defines four levels and §8.4.2 permits upgrade, downgrade

or termination after a conference; §5.3.7 calls for dual-mode communication under weak coverage in high-risk extreme weather—a conditional clause appearing once, not a blanket dual-channel mandate, and the communication-layer standard does not impose it. The BeiDou open-service specification (BDS-OS-PS-3.0) [6] reports $\geq 95\%$ success, sub-2 s average latency, an average service frequency of one message per 30 s and a 14,000-bit single-message ceiling, noting that realised frequency and maximum length are constrained by registration parameters: the rate is a provisioned quantity. These clauses bound lawful action—re-compile frequency within granted levels and request a conference change, never silently drop an obligation or declare risk.

The backup uses one 78-byte packet per 1200 s (20 B header, 58 B payload; a displacement sample is 6 B and rainfall 4 B, so roughly 9–10 samples fit). The elevated (yellow) task uses a 300 s period with a delivery deadline two periods after window open, a ≈ 600 s delivery window—shorter than the 1200 s backup slot spacing. The 200 B ceiling and ≥ 1 min period in DZ/T 0450 (§7.4.3.2, §7.4.2.3, with a ≥ 5 s inter-message gap at §7.4.3.5) belong to different service-formulation sections and are not combined into one “BeiDou operating point”; the 1200 s/78 B point is a declared research choice.

III. SYSTEM MODEL

Time is slotted at $\Delta=60$ s and $t=0$ is 06:00 local, so $t=12$ h is sunset and the solar cycle is explicit. Nodes $\mathcal{N}$, measurands $\mathcal{M}$ and a gateway g compose the field; a centre c lies beyond the backhaul.

A. Obligations and the delivery chain

A monitoring obligation is $o = (n, m, [l_o, h_o], d_o)$ over node n , measurand m , sampling window and absolute delivery deadline. A routine obligation at period P has $h_o - l_o = P$ and $d_o = h_o + P$, one period of grace. A sample fulfils o only if it matches (n, m) , is taken in-window and is received centrally by d_o , traversing

$$\underbrace{\text{qualified sample}}_{\text{node}} \rightarrow \underbrace{\text{heard at } g}_{\text{LoRa access}} \rightarrow \underbrace{\text{returned by } d_o}_{\text{cellular / short message}} \rightarrow \underbrace{\text{central evidence}}_{\text{fulfilled}}. \quad (1)$$

For piecewise missions the first segment uses a closed interval and later segments half-open windows, so a boundary sample cannot double-cover adjacent windows after a level change.

B. Links, outages, and who can act

LoRa access (node $\rightarrow$ gateway, with Class A downlink windows) succeeds per opportunity with probability $p_a=0.74$; the primary backhaul (gateway $\rightarrow$ centre) is good with probability $p_b=0.62$ outside a declared outage. A control outage is an interval $\mathcal{O} = [a, b)$ during which the primary is unavailable to centre $\rightarrow$ field traffic: centre commands and mission-table segments are refused at admission and never transmitted, and the short-message backup cannot carry downlink configuration. During $\mathcal{O}$ the only entities able to change field communication behaviour are the *node* (on its clock, own battery, cache,

recent access receipts and bounds carried by already-applied commands) and the *gateway* (on samples it has actually heard, its forwarding log and backup slots used). The centre retains authority to authorise changes but no channel to install them; a revision authored in $\mathcal{O}$ reaches the field at b .

C. Persistent installed state

Record state. Each unacknowledged sample s at node n has creation time t_s , a business deadline $d(s) = (\lfloor t_s/P \rfloor + 2)P$ under the locally known cadence, and occupies one of $K=240$ cache slots and up to $B=32$ positions in an access batch. Under the modelled edge discipline it leaves the cache only on central acknowledgement (modelled synchronously without airtime, equally favouring every queue rule); until then it is re-sent oldest-first.

Configuration state. The node runs one profile $q(t) \in \{P_0, P_1\}$ (sparse $P_0=600$ s, dense $P_1=300$ s) for both sampling and reporting, installed by an applied command carrying a version. A command is *applied* only when a later node report confirms the new profile (Class A latency); queued or in-transit commands change nothing. While applied, $q=P_1$ draws additional sampling and uplink energy every tick until something terminates it.

D. Energy

A sample costs $e_s=4.7 \times 10^{-4}$ Wh and an uplink $e_u=2.33 \times 10^{-5}$ Wh; battery capacity is $C=0.05$ Wh. Solar input is a 12 h half-sine with peak η (nominally 0.03 Wh/h) plus per-node/hour stochastic cloud occlusion. A node whose state of charge reaches zero *permanently* dies in the main configuration. Holding dense over a 12 h night draws ≈ 0.071 Wh $> C$.

E. Release time

For each state x , $t_{\text{rel}}(x)$ is the earliest time after which keeping x alive can no longer contribute to a fulfilled obligation. For a record, $t_{\text{rel}}(s) = d(s)$: past the business deadline the record cannot fulfil any obligation, so further retention or retransmission has zero delivery value. For a configuration profile, $t_{\text{rel}}(q)$ is the last deadline of an obligation the profile’s data can still meet through *some* feasible path,

$$t_{\text{rel}}(q) = \max_{o: \text{profile } q \text{ produces } o} \min\{d(o), \tau_{\text{last slot}}(o)\}, \quad (2)$$

where $\tau_{\text{last slot}}(o)$ is the latest primary or backup return opportunity at which a qualified sample for o can be heard and forwarded by $d(o)$. Unlike $d(s)$, this bound combines mission timing—which obligations the current profile serves—with return-slot geometry, when they can be carried; neither is fully observable at the node during $\mathcal{O}$.

F. Service loss and statement discipline

The objective counts deadline-valid obligation delivery (service), permanent node deaths, and the resource cost of state outliving its release time: wasted samples, access attempts, short-message bytes and energy. Offline, each unfulfilled obligation is charged to its *first* failing stage under an optimistic

model granting every later assumption, giving four mutually exclusive segments: S_{energy} (no qualified sample), S_{access} (sampled but not heard at the gateway by the deadline), S_{time} (heard but no return opportunity exists) and S_{cap} (an opportunity exists but capacity or congestion cannot carry it). An online observer that cannot see a stage reports *unknown* and never guesses it. Every feasibility claim is **confirmed** (an independent report shows the target config, or a matching sample is received centrally), **unconfirmed** (a command is in flight or receipts are incomplete, which implies neither that it took effect nor that the channel is down) or **unreachable** (the current path is known down from explicit network-side state). A LoRa access receipt—“a node was heard”—is evidence about the access link only and never by itself establishes backhaul reachability.

IV. PROBLEM FORMULATION

During an outage $\mathcal{O}$, at each tick t each acting entity chooses, for the state it can touch, an action: for a record $a_t(s) \in \{\text{RETAIN}, \text{RELEASE}\}$; for the configuration $a_t(q) \in \{\text{HOLD}, \text{REVERT}\}$, reverting dense to the last confirmed sparse profile. Let $\mathcal{E}_e(t)$ be the evidence entity e lawfully holds at t —for a node, clock, own SoC, cache contents, recent Class A receipts and any bound carried by commands applied before a , and explicitly *not* future mission-table segments, simulator-wide samples or backhaul truth. The local termination problem is

$$\begin{aligned} \textbf{Problem 1.} \quad & \min_{\{a_t\}} \quad \sum_o \mathbf{1}[o \text{ unfulfilled}] \\ & + \lambda_d \sum_n \mathbf{1}[n \text{ dies}] \\ & + \lambda_w \sum_{x: t \geq t_{\text{rel}}(x)} \text{cost}(x \text{ kept}) \quad (3) \\ \text{s.t.} \quad & \text{(C1) RELEASE/REVERT} \Rightarrow t \geq t_{\text{rel}}(x), \quad (4) \\ & \text{(C2) SoC}_n(t) > 0 \quad \forall n, t, \quad (5) \\ & \text{(C3) } a_t = f(\mathcal{E}_e(t)), \quad (6) \\ & \text{(C4) unknown} \Rightarrow \text{unconfirmed/unfulfillable}, \quad (7) \end{aligned}$$

where (C1) is no false release, (C2) energy survival, (C3) evidence locality and (C4) no false completion, which forbids reporting either fulfilment or unreachability under unknown feasibility. The tension is between (4) and (5) under locality (6). Releasing too early violates (4), abandoning work that could still be delivered; releasing too late violates (5), paying for dense data after they became undeliverable and discharging fatally. The difficulty is that the two state types observe t_{rel} very differently.

A. Record sub-problem: t_{rel} is locally certain

Because $d(s)$ uses only the public cadence, the node evaluates $t_{\text{rel}}(s) = d(s)$ with certainty and no downlink. Constraint (4) is the exact test $d(s) \leq t$, and RELEASE frees both the backup send and the access batch. The optimal local rule is ordinary per-record expiry; §V-A shows a certificate-keyed variant is the same rule, and that the only design choice with measurable consequence is *where the release is executed*, through (6) and the cross-segment acknowledgement coupling.

B. Configuration sub-problem: t_{rel} is not locally observable

During $\mathcal{O}$ the node cannot compute $t_{\text{rel}}(q)$ from (2): mission revisions that change which obligations q serves travel the failed backhaul, and return-slot geometry sits at the gateway. Three local approximations are available, each flawed. A *clock guard* reverts at sunset, correct when $\mathcal{O}$ aligns with dusk and dense data are undeliverable by then, and blind to a daytime downgrade and to overcast days with little harvest. A *fixed config TTL* reverts Δ after install, and a single Δ cannot track (2): short Δ violates (4) when the dense window is long, long Δ violates (5) under tight energy when the window is short. A *local energy guard* reverts on a SoC threshold computed at the node; it is necessary and deployed as a safety net, but reactive, acting only once energy is already low and unable to tell whether the data being sacrificed were still deliverable.

Delivery-bounded lease. While connected, the gateway computes (2) for the profile it is about to install, using obligations it knows and the primary/backup slot schedule, and attaches the resulting absolute bound τ_L to the configuration command. The node stores τ_L at apply time and reverts locally at $t \geq \tau_L$ with no downlink. The bound uses only information valid at install time and never forecasts a future authorisation; a later connected revision arrives with a fresh lease. This is the ordinary config-lease/TTL primitive with its bound re-derived from delivery geometry, exactly as semantic HARQ reuses HARQ but changes its criterion [2]: the minimal local object that makes (4)–(5) jointly satisfiable without assuming a surviving control path. When even the lease cannot keep the node alive, the honest outcome under (7) is to revert to the safe sparse default and mark the interval’s obligations unfulfillable, not to report them complete.

C. Why this is a communication problem

Problem 1 is defined over network state, evidence locality and return geometry; it has an optimal oracle solution and ordinary rule-based solutions independent of any LLM. An agent is one possible *solver* of the same decision and a convenient probe of where the interface misleads a planner; nothing in the objective or constraints references a model. We therefore evaluate the mechanisms deterministically and treat the LLM study as formative.

V. MECHANISMS

A. Record termination: deadline-aligned expiry

At each upload opportunity the node deletes cached records with $d(s) \leq t$ and sends survivors oldest-first. We implement two formulations: *deadline-purge*, keyed to the obligation cadence, and *generic-expiry*, an ordinary BPv7-style per-record lifetime assigned at creation, expires = $t_s + ((P - t_s \bmod P) + P) = d(s)$, referencing no obligation id, slot geometry or certificate. A gateway variant locally acknowledges records it will not send. EDF and obligation-greedy queues only reorder and never release; AoI latest-only releases indiscriminately, keeping only the newest. Dropping stale or expired traffic is well established in AoI packet management and DTN bundle expiry [7], [8],

[9], [3], and suppressing sensor traffic whose delay bound is exceeded is likewise prior art [10]. We claim no new discard algorithm. The system-level content is the end-to-end business-deadline basis—not packet age, hop TTL or an estimated single-packet delay—zero-downlink source placement, and, decisively, cross-segment consistency, because the acknowledgement coupling means a gateway-only release does not free the source cache.

B. Configuration termination: guard and delivery-bounded lease

The *local clock/energy guard* is mature: on every tick the node, on its local clock, clamps sampling and reporting to sparse at night ($\text{hod} < 06:00$ or $\geq 18:00$) and leaves daytime commands untouched, and an optional computed energy ledger—remaining dark hours times the dense-versus-sparse extra draw, against the node’s own SoC—additionally reverts when dense operation would not survive the night. This is a standard time/energy local rule modelling the gateway’s pre-provisioned autonomous rules; it is off by default and bit-identical when off.

The *delivery-bounded lease* is the residual mechanism. The gateway computes τ_L (Eq. 2) while connected: for the current dense profile it takes the last obligation the profile produces before the next known revision, sets that obligation’s qualified-sample and return-slot deadline, aligns to the latest feasible primary or backup slot, and attaches τ_L to the install command. The node enforces revert at τ_L alongside the clock guard, which remains the always-on safety net for conditions the install-time bound cannot see, such as an unforeseen weather collapse. A pre-provisioned full schedule is the degenerate case in which τ_L for every segment is known at $t=0$ and a local schedule executor suffices with zero centre traffic.

C. Partial-observability projector

At compile time the centre holds the obligation table, the *current* and never forecast network-side outage state, and public parameters; at mission-table arrival the gateway holds only what it has heard. A structural no-return certificate (S_{time}) is emitted from slot geometry alone and is explicitly conditional on the primary being unavailable; geometry never infers an outage. An arrival-time projector labels a missed obligation only when local evidence separates the stages, and otherwise returns unknown. An unknown-aware centre projector emits *unreachable* solely from an explicit network-side field—which the current harness does not yet provide, so it emits *unconfirmed*—and replaces any blanket night kill with the computed energy ledger, so genuine energy-infeasibility is distinguished from conservative advice.

VI. PROTOTYPE AND METHODOLOGY

All mechanisms run in one Python event-stepped simulator (60 s ticks) implementing LoRaWAN Class A downlink windows, versioned two-field configuration commands (sampling interval and reporting period delivered as independent fields, with cross-generation mixed configs detected), the hold-until-acknowledged edge cache, max-coverage backup packing,

TABLE I
COUNTERFACTUAL INTERVENTIONS OVER 7560 OBLIGATIONS
(EPISODE I). SET DECOMPOSITION OVER THE TESTED CONFIGURATIONS;
NOT AN UPPER BOUND OVER ALL LEGAL POLICIES.

Run	Sampling/Energy	Backhaul	Delivered
base	dayfeed, 0.05 Wh	1200 s/78 B	3025 (0.400)
O_{bh}	dayfeed, 0.05 Wh	unlimited	4740 (0.627)
O_{en}	always-300, 0.50 Wh	1200 s/78 B	4089 (0.541)
O_{bo}	always-300, 0.50 Wh	unlimited	6023 (0.797)

stochastic solar and cloud, absorbing brownout, and the non-prescient piecewise mission-view gate. Disabling the backup reproduces the v1.1 simulator bit-for-bit across five anchored invariants; the local guard and lease are default-off.

Two controlled episodes are used. *Episode I*, the v1.1 manifest point, upgrades to yellow at $t=6$ h with primary outage 4–20 h and the mission table reaching the gateway at 20 h; it drives the frontier, attribution, record-expiry, placement and live-agent results. *Episode II* is a deterministic configuration-termination scenario with identical parameters except schedule, outage phasing and swept harvest: to place a daytime downgrade *inside* an outage, phase A uses yellow at $t=2$ h with a conferred downgrade at $t=6$ h and outage 4–20 h, phase B uses yellow at $t=1$ h, downgrade at $t=8$ h and outage 6–22 h; $t=12$ h is sunset. Episode II is an explicitly constructed controlled comparison for the configuration subproblem, not a new field claim. Node deaths, deterministic and highly robust, are reported as totals over seeds 0–2; sub-point mean-service differences at $n=3$ are treated as not significant; seeds 6–9 are a reused held-out confirmation set, not an untouched test set. Every figure is regenerated by a registered repository script and the suite reports 20/20 self-checks.

VII. EVALUATION

A. Where the fixed-resource loss is

Over Episode I’s 7560 routine obligations the strong max-cov/FIFO baseline delivers 3025 (0.400). Symmetric per-obligation counterfactuals widen one physical resource at a time, never the policy, and report set differences over identical obligation ids. Relaxing backhaul to unlimited gives 4740 (0.627), relaxing energy (0.50 Wh, always-dense) gives 4089 (0.541), and both gives 6023 (0.797); 1715 obligations are blocked by backhaul capacity alone, 1128 by battery alone, and 155 (2.0%) by both (Table I). These are intervention results on implemented strategies, not an exhaustive optimum; we make no zero-remainder or global-optimality claim. They establish that the large throughput gains are physical, which is why we then target *state termination* rather than a smarter scheduler.

Charging each of the 4535 failures to its first failing segment, with “heard” required by the *deadline* rather than at any later time, gives S_{energy} 2002 (44.1%), S_{cap} 1115 (24.6%), S_{access} 830 (18.3%) and S_{time} 588 (13.0%). An earlier heard-ever accounting had reported S_{cap} 1945 and S_{access} 0: all 830 moved obligations were in fact sampled but first heard at the gateway after their deadline, late access rather than backhaul

TABLE II

TEN-SEED EPISODE I SWEEP WITH THE STRONG MAXCOV CHOOSER FIXED; ONLY THE EDGE DISPOSITION CHANGES. GATEWAY-ONLY SUPPRESSION (NOT SHOWN; SAME FIFO CACHE) GIVES 194 OUTAGE ON-TIME AT SEED 0, AGAINST 202 FOR FIFO AND 434 FOR SOURCE EXPIRY—THE CROSS-SEGMENT COST. [†]STANDARD BPV7-STYLE PER-RECORD LIFETIME; [‡]OBLIGATION-KEYED PURGE, BIT-IDENTICAL TO [†].

Edge disposition	Full svc	Outage OT	Expired	Deaths
FIFO (baseline)	0.373	1691	2538	12
latest-only (AoI)	0.358	4250	0	0
generic-expiry [†]	0.415	4265	1	0
deadline-purge [‡]	0.415	4265	1	0

capacity. The delivered count (3025) is identical under both accountings; the correction changes *where the loss is charged*, and therefore which mechanism could address it.

B. Record expiry: standard primitive, decisive placement

During the Episode I outage 75 backup packets carry 750 records; under FIFO 219 arrive centrally already expired against 261 on-time (45.6%; 1270 of 2782 payload bytes). Gateway-only suppression removes those sends but withholds the acknowledgement, and on-time delivery falls 202 $\rightarrow$ 194: cost moves to the access segment. Source-local deadline expiry reverses the sign, raising outage on-time delivery 202 $\rightarrow$ 434 and the deadline-valid fraction 0.54 $\rightarrow$ 1.00. Across ten seeds (Table II) expiry raises mean service 0.373 $\rightarrow$ 0.415, a paired +4.21 points (95% CI +3.64 to +4.79, positive on all ten), total outage on-time delivery 1691 $\rightarrow$ 4265 (2.52 $\times$), expired backup records 2538 $\rightarrow$ 1, and deaths 12 $\rightarrow$ 0; it beats AoI latest-only by +5.73 points (CI +5.06 to +6.39), because latest-only buys outage freshness by abandoning post-recovery coverage. Crucially, deadline-purge and generic-expiry are bit-identical on every seed—service, outage counts, deaths and delivered set—so the record mechanism *is* standard per-record expiration, and its measured value is the business-deadline basis plus source/gateway cross-segment consistency, not a new discard algorithm.

C. Configuration leases: deaths under overcast skies, false release under short TTL

Holding record expiry fixed (generic-expiry with maxcov) and varying only configuration termination in Episode II, the node profile trace confirms the clock guard holds dense until sunset at $t=12$ h: 12 samples/h in [2, 12) h and 6/h thereafter. In phase A the last yellow obligation is released just before the noon downgrade, and its deadline aligned to a 1200 s backup slot gives $\tau_L=6.33$ h. Sweeping the local revert time (Table III), reverting at $\tau_L=6$ h removes 468 of 978 (48%) post-deadline daytime dense samples with *zero* loss of yellow delivery (356/672) and no deaths, while reverting as early as 5 h falsely releases 9 yellow obligations, marking the no-false-release boundary. At nominal harvest ($\eta=0.03$) service, deaths and final SoC are unchanged: here the lease buys resource and airtime efficiency, not service.

Sweeping harvest over three seeds, $\eta=0.03$, 0.02 and 0.015 give 0 deaths and service ≈ 0.597 under every bound. At

the overcast $\eta=0.012$ the clock guard’s sunset lag kills 20 nodes (service 0.5961, final SoC 0.066) while the lease at 6–8 h removes all 20 deaths (0.5970, SoC 0.122, near-doubled); reverting as late as 10 h also avoids deaths but at a smaller margin (SoC 0.117). At $\eta=0.01$ even sparse operation cannot be powered (≈ 40 deaths under every bound); the lease modestly raises pre-death service (0.530 $\rightarrow$ 0.577) but cannot save nodes, and the correct output is an unfulfillable declaration rather than a false completion.

Phase B needs the bound near 8.3 h, the long dense window, where a 4 h TTL relaxes at 5 h and falsely releases 183 yellow obligations (service -1.1 points) and a 6 h TTL still loses 26; phase A needs it near 6.3 h, where an 8 h TTL only just survives at the tightest energy (SoC 0.117). The same pattern holds at nominal harvest. The delivery bound attains each phase’s optimum because it is computed from obligations and slots rather than guessed as a duration. That is the mechanism’s measured increment over a plain config TTL: phase- and rate-adaptive termination under (4)–(5), with a bound that is auditable—“stop because the data cease to be deliverable here”—rather than empirical. We report it as a point-specific, deterministic result over the tested phases; wider sweeps of backup rate and node count are future work.

D. What the field can attribute when the table arrives

In Episode I the delayed mission table reaches the gateway at $t_a=20$ h, by which point 2136 yellow obligations are past deadline. Offline time-aware attribution labels all 2136 correctly (S_{time} 588, S_{cap} 327, S_{access} 650, S_{energy} 571). Using only evidence the gateway lawfully holds by t_a labels 1138/2136 (53.3%)—all geometric S_{time} (588), all in-time-not-returned S_{cap} (327) and 223 provably late S_{access} —at 100% precision on the labelled subset; 998 (46.7%) remain unknown, whose offline truth is 571 never-sampled and 427 heard only after t_a (Table IV). “Never sampled” and “sampled but not yet heard” are inseparable without evidence the gateway does not have, and must stay unknown, which is the original G5 constraint. Compile-time geometry, needing no sample, emits 602 conditional no-return certificates of which 588 are true S_{time} , the 14 recovery-edge cases being excluded rather than certified; an optimistic-energy capacity certificate is impure (552 correct S_{cap} , 571 actually S_{energy} , 193 false positives), so capacity is left to the gateway. The design consequence is exactly Problem 1’s: local record and configuration action must not wait on resolving these unknowns.

E. Enforcement placement determines survival

A centre-side feasibility *envelope*—a pure filter over any planner using only confirmed configs, hearing counts, the authorised requirement and the clock—removes the command storm but cannot save the tightest seed, because a dusk downgrade must cross the backhaul and a Class A window it misses. The same “never stay dense overnight” rule is a late, refusable suggestion at the centre and an invariant enforced every tick at the node (Table V). Naive centre compliance kills 13 of 14 nodes on all three seeds (39 total) at service 0.162; adding the node-local guard removes all deaths at unchanged

TABLE III

CONFIGURATION TERMINATION ACROSS TWO OUTAGE PHASES (SEEDS 0–2, GENERIC-EXPIRY WITH MAXCOV FIXED). THE CLOCK GUARD LEAVES 20–31 OVERCAST-SKY DEATHS. NO FIXED TTL IS SAFE IN BOTH PHASES: 4 h FALSELY RELEASES 183 STILL-DELIVERABLE OBLIGATIONS IN PHASE B, AND 8 h IS THE PHASE B VALUE WHILE LEAVING PHASE A AT MINIMAL ENERGY MARGIN. THE DELIVERY BOUND TRACKS EACH PHASE’S LAST DELIVERABLE SLOT, REACHING 0 DEATHS WITH NO FALSE RELEASE AND THE HIGHEST FINAL SoC. AT $\eta=0.03$ ALL ARMS HAVE 0 DEATHS AND THE LEASE CHANGES ONLY WASTED DENSE WORK; AT $\eta=0.01$ EVEN SPARSE OPERATION IS INFEASIBLE (ALL ARMS ≈ 40 DEATHS), WHICH THE SYSTEM MUST REPORT AS UNFULFILLABLE RATHER THAN SURVIVE.

Local revert bound	Deaths (3 seeds)	Mean svc	Mean final SoC	Yellow delivered
<i>Phase A, overcast $\eta=0.012$ (upgrade 2 h, downgrade 6 h, outage 4–20 h)</i>				
clock guard (sunset 12 h)	20	0.5961	0.066	849
fixed TTL 4 h (revert 6 h)	0	0.5970	0.122	849
fixed TTL 6 h (revert 8 h)	0	0.5970	0.122	849
fixed TTL 8 h (revert 10 h)	0	0.5971	0.117	849
delivery lease ($\tau_L=6.33$ h)	0	0.5970	0.122	849
<i>Phase B, overcast $\eta=0.012$ (upgrade 1 h, downgrade 8 h, outage 6–22 h)</i>				
clock guard (sunset 12 h)	31	0.5981	0.038	2020
fixed TTL 4 h (revert 5 h)	0	0.5881	0.122	1837
fixed TTL 6 h (revert 7 h)	0	0.5994	0.122	1994
fixed TTL 8 h (revert 9 h)	0	0.5996	0.122	2020
delivery lease ($\tau_L=8.33$ h)	0	0.5996	0.122	2020

TABLE IV

ARRIVAL-TIME ATTRIBUTION OF THE 2136 UPGRADED OBLIGATIONS ALREADY MISSED WHEN THEIR TABLE REACHES THE GATEWAY (EPISODE I). THE ONLINE PROJECTOR IS EXACT BUT PARTIAL; 427 MATCHING SAMPLES ARE FIRST HEARD AFTER t_a AND 650 AFTER THEIR DEADLINE.

Segment	Offline truth	Online at t_a
S_{time} (no return slot; geometry)	588	588
S_{cap} (heard by deadline, not returned)	327	327
S_{access} (heard after deadline)	650	223
S_{energy} (no qualified sample)	571	— (unknown)
U (not separable online)	—	998
Total	2136	2136
Labelled coverage	100%	53.3%
Precision on labelled subset	1.000	1.000

service; with the guard present the envelope additionally cuts refused admission attempts to 222. A zero-command pre-provisioned rhythm reaches 0.360 service, matching every centre-compiled arm for these static missions—a scoped negative result on centre sampling control for pre-provisioned tasks, which is exactly the gap the delivery lease targets for *non*-pre-provisioned runtime changes.

F. Formative real-agent study

We ran deepseek-flash live over the Episode I outage: three seeds each of an unaided agent (A0), the agent with an ordinary pending-command checklist (A0s) and a certificate agent (A1), plus two healthy-link controls, with a real model call at each 30-minute decision, 99 per trace and 1089 decisions in total; all arms share harness, legal observation, tools, history and action space. The study motivates and stress-tests the interface and is not used to establish the deterministic mechanisms.

Service and survival, noisy at $n=3$. A1 attains mean service 0.358, level with the 0.360 deterministic safe floor, with 4 deaths; A0 0.296 with 35 deaths and A0s 0.222 with 39; the

healthy-link controls give A0 0.275/13 and A1 0.511/0. Per-seed service overlaps, so we draw no service significance. The live agent loop was not wired to the local guard, so the four A1 seed-1 deaths are not cleared by the deterministic result of §VII-E.

Call accounting. The 1089 decisions are not 1089 successful calls: 52 ended in a parse-failure hold from `max_tokens` cutting the JSON mid-string, and 412 records carry a first-attempt error even when the retry succeeded, a logging artefact. Per-arm parse-failure rates are A0 7.1% (28/396), A0s 6.4% (19/297) and A1 1.3% (5/396), so arm differences mix output-format reliability with any semantic effect.

The interface fault, symmetrically scored. The observation’s link field carries only LoRa access receipts (`n_heard`) and no backhaul state. In the forced outage window A0/A0s assert “link up, commands can apply” on 36/32 decisions, and A1 on 1 after token-boundary de-noising of an “uplink-only” substring—deterministically over-optimistic, since the primary is down throughout, and accompanying the ineffective dense orders and deaths. In the opposite direction the v4 certificate asserted “control plane not delivering” whenever fewer than all nodes were heard, firing at five healthy-link times when the primary was in fact usable, twice after 13 of 14 nodes had already confirmed dense. The earlier false-report tally was invalid on three counts: it ran only on decisions that ordered dense, vacuous for the mostly-hold A1; it never ran on the healthy group, so those zeros were initialisation; and it counted keywords the certificate prompt itself taught, terms appearing in 317 A1 decisions and zero A0/A0s decisions. Offline replay of an unknown-aware v5 projector on identical snapshots issues zero unreachable assertions without a network-side field, turning all five healthy false alarms into unconfirmed, and splits 528 blanket night kills into 189 genuinely energy-infeasible and 339 advisory, every near-dawn 04:30–06:00 case being advisory. The honest finding is that a missing backhaul-state signal causes bidirectional errors; isolating any independent increment of the projector requires a matched-

TABLE V

DETERMINISTIC EPISODE I PLACEMENT STUDY, NO LLM. THE LOCAL GUARD REMOVES ALL PERMANENT DEATHS IN THE TESTED SEEDS AT UNCHANGED SERVICE; THE CENTRE ENVELOPE ALONE CANNOT. REFUSED COMMANDS ARE REJECTED AT ADMISSION AND NEVER TRANSMITTED, WHICH IS COMMAND HYGIENE RATHER THAN BILLING SAVING. POINT-SPECIFIC EMPIRICAL RESULTS, NOT A SURVIVAL INVARIANT OR OPTIMAL BOUND.

Configuration	Mean svc	Deaths	Cmds sent	Refused
comply (naive, unprotected)	0.162	39	160	2488
dayfeed-c (centre day/night rule)	0.370	12	308	1256
envelope(comply), centre only	0.359	12	308	298
comply + local guard	0.366	0	1424	3364
dayfeed-c + local guard	0.368	0	258	1156
envelope(comply) + local guard	0.359	0	282	222
pure-local (pre-provisioned, 0 cmds)	0.360	0	0	0

capability arm receiving the same constants and heuristics without the projector, end-to-end v5 runs with an explicit backhaul-state field, and cross-model replication, all future work.

VIII. RELATED WORK

LLMs in emergency and network control. TopoLLM applies adaptive tool learning to post-disaster topology planning [11], and later systems let LLMs schedule UAV paths and bandwidth, shape RL rewards or orchestrate recovery tools, including intent objects, budget and time constraints, candidate-plan checks, local repair and documented network-management APIs [12], [13]. We claim no novelty for natural language to constraints, nor for tool orchestration. These decide network resources or compose recovery tools under an assumed reachable control path; they do not terminate state that must keep acting after that path dies, and do not maintain an unknown-aware three-class state over a long routine monitoring workflow. Agentic Wi-Fi likewise uses an LLM as a solver for a separately defined coordination protocol; we follow that ordering—problem first, agent optional.

Scheduling, freshness, energy. The closest state-model precedent is an LLM in-context data-collection scheduler with bounded per-sensor queues, exogenous arrivals, finite service opportunities and feedback [14]; AoI and energy-harvesting control, including instantaneous-battery threshold policies and partially observed harvesting, address resource pieces [15], [16], [7]; overbooking and contact-aware forwarding for space links [17] and network-calculus bounds [18] are complementary deterministic characterisations. Joint sampling, transmission and energy optimisation is established and not claimed here; our distinction is persistent *remote* configuration with Class A apply latency, a separated access versus primary/backup return path, and termination under a control outage. Consistent network update research has long studied per-update validity, dependencies and safe transitions [19]; we rely on that lesson and add the disconnected, energy-constrained edge case in which an installed update's *own data* become undeliverable before a counter-update can arrive.

Packet discard, DTN expiry, contacts. AoI management pre-empts or drops old updates [8], [20]; deadline-aware buffer discard has been coupled with adaptive ARQ for video [2]; DTN bundle protocols attach creation-time lifetimes and use

TTL or encounter drop [9], [21], [3], and custody transfer [22] moves retention responsibility as a forward hand-off, since migrated out of the BPv7 core. Rural LoRa–LEO systems schedule around estimated satellite contacts and exploit spatial gateway diversity [1]. Our record disposition is BPv7-style expiry and we prove the equivalence; our configuration lease is a TTL/lease primitive with a delivery-derived bound. The claimed deltas are narrow and explicit: an end-to-end business-deadline basis with cross-segment source/gateway consistency, since expired backhaul suppression alone backpressures access, and a configuration bound computed from the last deliverable slot, so that termination is safe across outage phases without a live control path.

IX. DISCUSSION, LIMITATIONS, AND THREATS TO VALIDITY

Scope. The throughput negative result and the resource walls are scoped to the policy families and one operating point; they are not an optimum, and a better scheduler could exist. The lease result is deterministic over two constructed phases and five harvest levels, establishing the mechanism's existence and the failure of fixed clock and TTL bounds across those phases, not a universal parameter choice. Wider sweeps of backup rate and payload, node count, capacity, cloud model, restart thresholds and non-absorbing brownout are future work and could move the energy wall, not the slot or capacity walls. The lease bound uses information valid at install time; an authorisation changed entirely during a long outage is physically unreachable, and the correct behaviour is safe-default revert plus an unfulfillable declaration, which we report rather than hide.

Model simplifications. Central acknowledgement is synchronous and airtime-free, equally favouring every queue rule; gateway acknowledgements would consume Class A windows in deployment. During Episode I the node tests deadlines against the public pre-change cadence, with post-upgrade deadlines used only in offline accounting. Single fixed-period missions make obligation-keyed and deadline-keyed record release coincide; multi-obligation or overlapping mission versions require release keyed to a record's latest still-valid deadline. Episode II is a controlled scenario for the configuration sub-problem, not an independent field measurement.

Agent evidence. The live study is single-model, three seeds per arm, temperature 0, with large per-run variance and a confounded expert packet; it demonstrates the interface fault and the projector’s offline effect, not an independent agent increment. The hardened harness—per-request accounting, symmetric scoring, unknown-aware projector and energy ledger, default-off local guard and lease—is in the repository so that matched-capability, end-to-end v5 and cross-model studies can run without re-litigating instrumentation; those are explicitly not claimed here.

Provenance. One standards channel was obtained from a document-sharing host, a top-grade document with a lower-grade copy chain, because the official host was unreachable by TLS; clause numbers were cross-checked from two independent copies, and unverified international LoRa-plus-satellite and vendor lines are not used as fact. We did not execute cited systems’ implementations and claim no replication of them.

X. CONCLUSION

After a backhaul failure the communication state installed beforehand keeps executing, and its termination cannot wait for the control path that failed. We formulated this as one local termination problem over two kinds of persistent installed state. Records carry a locally certain release time, the business deadline, and standard per-record expiry placed consistently at the source doubles outage on-time delivery while gateway-only suppression merely moves cost to the access segment. Configurations carry a release time the node cannot observe during an outage; a fixed clock guard or fixed TTL either kills nodes under overcast skies or falsely releases deliverable obligations across phases, whereas a delivery-bounded lease—an ordinary lease whose bound is the last deliverable slot, installed while connected and enforced locally—removes those deaths without false release. Throughout, mechanisms act only on evidence the acting entity lawfully holds, say unknown when they cannot tell, and declare work unfulfillable rather than complete. The network problem and its deterministic solutions stand on their own; an LLM is one candidate controller, and our live study shows most clearly the interface fault—access receipts mistaken for backhaul state—that the evidence discipline is designed to prevent.

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